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Explainable Federated Learning for Healthcare Time-Series Forecasting with Personalized Drift Adaptation (FPDAF)

23CSE399 – Project Phase 1 (Technical Progress Report - Review 2)
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1 Project Proposal and objectives

Overall Project Goal: Instead of merely implementing standard off-the-shelf algorithms, this project sets the objective to develop a novel extension to existing FedAvg and FedPer models. We will introduce the fully decentralized Federated Personalized Drift-Aware Attention Framework (FPDAF) specifically designed for multivariate healthcare time-series analysis (e.g., ICU risk forecasting).

Primary Novelty & Proposed Algorithmic Innovation: The definitive novelty of FPDAF lies in its planned hybrid integration. Unlike conventional FL architectures that maintain blind static global averaging, our extension proposes to intercept the FedAvg loop. We will inject a Novel CUSUM Extension directly into local active hospital nodes, enabling isolated Neural Network layers to act as an Automated Neural Trigger. This allows the model to "unfreeze" and adapt in real-time when patient physiology rapidly deteriorates, utilizing Temporal Attention to explain exactly which vitals caused the deviation.

Key Research Pillars:

- Novel FedAvg Extension: Uses Attention-Aware Weighting to prioritize higher-confidence hospital data during the global update instead of just simple patient-count averages.
- Novel CUSUM Extension: Repurposes statistical monitoring as an active neural manipulator for instant clinical adaptation.
- Novel Local Personalization: Extends standard FedPer nodes to focus on specific hospital demographics without losing shared global intelligence.

2 Introduction and Background

Clinical Context & Terminology:

- Sepsis: Life-threatening organ failure caused by infection; needs early detection.
- Deterioration: Rapid worsening of patient condition (vitals becoming abnormal).
- Profiling: Analyzing time-series data (Heart Rate, BP) to predict patient risk.

Continuous monitoring systems in Intensive Care Units (ICUs) collect immense streams of physiological data (HR, BP, SpO2). Early detection of patient deterioration using Deep Learning models

can prevent fatalities. However, massive clinical models require centralized data, violating privacy laws. While Federated Learning (e.g., FedAvg) resolves the data-sharing issue, it often fails to account for hospital-specific demographics and protocols (Accuracy Gap) or lacks clinician trust (Trust Gap).

Sustainable Development Goal (SDG) Mapping:

- Primary (SDG 3): Directly contributes to Good Health by enabling explainable patient risk prediction.
- Secondary (SDG 9 & 16): Supports scalable clinical innovation and ethical data privacy security.

3 Execution plan

The subdomains required involve: Data Engineering (custom ICU sequence windows), Deep Learning Base (extending LSTMs with novel Temporal Attention), Decentralization (Attention-Aware FedAvg), Customization (dynamic FedPer heads), and Continuous Adaptation (iterative feedback loops for real-time recalibration).

4 Fundamentals which you have learned in the semester

To prepare for implementation, the team completed an extensive curriculum:

- Textbooks & Papers: Thoroughly studied Federated Learning architectures (FedAvg, FedPer) to understand secure aggregation.
- PyTorch Engineering: Mastered custom training loops and gradient isolation techniques in PyTorch to separate "Global Backbones" from "Local Heads".
- The Privacy Gap: Studied international and local regulations—HIPAA and India's DPDPA 2023 (Section 8)—which mandate decentralized data handling.
- Statistical Mechanics: Learned the equations defining the CUSUM drift detection standard for monitoring temporal shifts in clinical residuals.

5 Architecture diagram and explanation of the design

The FPDAF architecture is a split-layer design as shown in Figure 1. It utilizes a centralized server for global weight aggregation while maintaining strictly isolated local personalization heads at each hospital node to ensure privacy and specialized clinical accuracy.

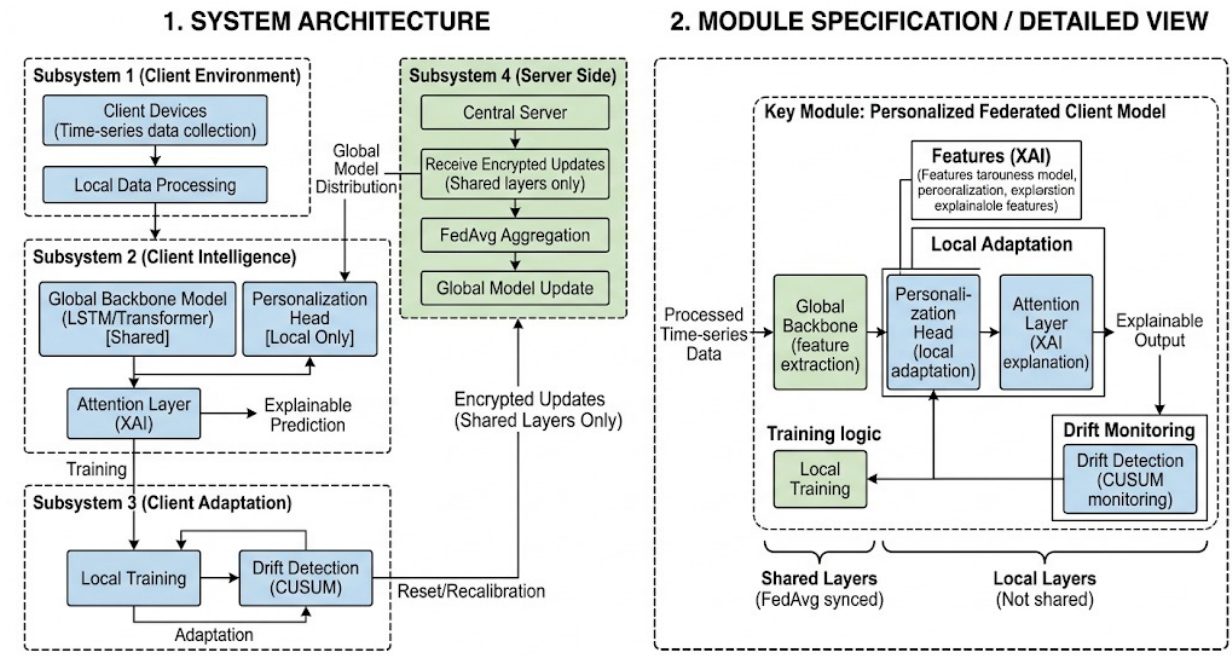

Figure 1: Master System Architecture: Highlighting the FPDAF extension for early ICU risk prediction.

The Attention Layer sits atop the backbone to compute temporal context vectors based on recent physiological spikes, giving transparent x_i timestamps feature importance. Finally, the Personalization Head is isolated completely to the hospital node; it never transfers its gradients to the server, ensuring the final clinical decision is strictly correlated to the local hospital's demographic distribution.

Comparative Framework Analysis:

Mechanism	Standard FedAvg	Standard FedPer	Proposed FPDAF
Aggregation	Full global averaging	Partial local-global split	Attention-Weighted Syncing
Drift Handling	Manual Retraining	Static Personalization	CUSUM Neural Trigger
Explainability	Black-box predictions	Local feature importance	Temporal Attention Weights
Personalization	None (One Global Model)	Multi-hospital site tuning	Dynamic Personalization

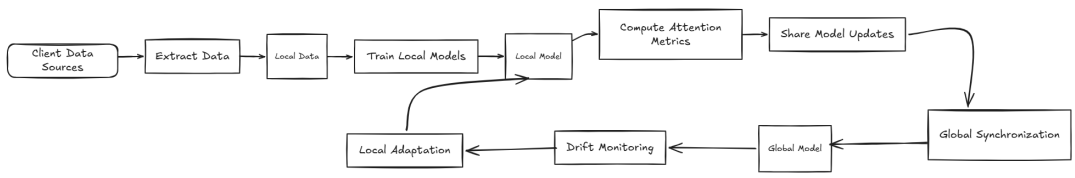

Figure 2: FPDAF System Dataflow Mapping - Illustrating end-to-end data processing.

6 Scientific Methodology & Neural Formalism

The innovation of FPDAF is formalized as a split-weight optimization problem across K hospital nodes. The local predictive output $\hat{y}_t^k$ is defined by:

$$\hat{y}_t^k = h_{\phi^k}\left(g_\theta(X_{1:t}^k) \cdot \alpha_t\right)$$

where g_θ is the Global Temporal Backbone (synchronized via FedAvg), h_{ϕ^k} is the Private Personalization Head (site-specific), and α_t represents the Temporal Attention context vector.

To handle temporal drift, we implement the CUSUM Neural Trigger. The system monitors the cumulative deviation of local prediction residuals:

$$S_t = \max(0, S_{t-1} + |y_t - \hat{y}_t| - \mu - k)$$

If $S_t > H$ (the statistical threshold), the framework automatically declares a patient state-drift, triggering a local backpropagation cycle on the personalization head ϕ^k while keeping θ frozen, preserving the global model's generalizability while adapting to the patient's acute deterioration.

7 Proposed Phase 2 & 3 Roadmap

The direct roadmap to full implementation is detailed in Table 2.

Target Area	Objective for Phase 2 & 3	Tentative Time
Data Preprocessing	Clean, Impute, and construct sequential windows for PhysioNet.	Ongoing – April 2026
Baseline Training	Code standard FedAvg on PyTorch locally for forecasting baselines.	April – June 2026
FPDAF Integration	Embed CUSUM triggers and Attention layers into personalized heads.	July – Sept 2026
Federated Simulation	Deploy 3-5 simulated Hospital "Clients" to test adaptation latency.	Oct – Dec 2026
Final Evaluation	Overall system performance benchmarking (Accuracy, F1-Score, AUC-ROC).	Jan – Feb 2027

Table 1: Project Execution Roadmap (2026-2027)

8 Summary of works completed in Phase 1

During this semester, the following milestones were successfully completed:

1. High-Era Dataset Aggregation: Secured a clinical stack including PhysioNet (2019, 2012, 2026), VitalDB, and EDT-Datasets.
2. Novelty Identification: Defined the FPDAF architectural extension fusing Federated Learning with proactive Drift Detection.
3. Mathematical Definition: Derived the formal split-weight optimization framework for localized personalization.
4. Data Engineering: Developed the initial Python sequence profiling script to handle ICU NaN issues and standardization.
5. Environment Setup: Established the official repository and verified standard FedAvg baseline feasibility.

Appendix: Data & Resource Access

Primary Project Repository: https://github.com/23CSE-TEAM-28/23CSE399-Project-Phase-1.git

Sepsis Challenge Data: https://physionet.org/content/challenge-2019/1.0.0/

PhysioNet 2012 Mortality: https://physionet.org/content/challenge-2012/1.0.0/

Trajectory Flow Matching (2026): https://physionet.org/content/clinical-trajectory-flow-icu/1.0.0/

VitalDB Dataset (2020): https://physionet.org/content/vitaldb/1.0.0/

EDT-Datasets (2021): https://github.com/mommand/EDT-Datasets

Full Literature Review (25 Papers): https://docs.google.com/spreadsheets/d/1G_zAbr6MI_bYcCHEXfkWjhmvKzBEDKr0/edit#gid=417550448

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